

The Utility Problem

Energy disaggregation, also known as appliance disaggregation is a technique which is used to analyze and break down the energy consumption in a building or household into individual appliance-level energy usages. The goal is to identify and monitor the energy consumption of specific appliances without the need for additional metering or sensors on each device.

The process of energy disaggregation involves analyzing the overall power signal from a building or household and applying advanced algorithms and machine learning techniques to separate and attribute energy consumption to specific appliances. This is typically achieved by examining distinctive patterns and signatures in the power signal that correspond to the operation of different appliances.

Energy disaggregation has several applications and benefits, including:

- Energy management - by understanding the energy usage of individual appliances, homeowners, facility managers or utility representatives can gain insights into energy consumption patterns, identify energy-hungry appliances, and make informed decisions about energy efficiency measures.
- Load profiling and demand response - utilities can use energy disaggregation to obtain detailed load profiles of their customers, helping them develop targeted demand response programs (like Time-of-Use rates) and optimize energy distribution.
- Billing and tariff optimization - by analyzing energy consumption some of the specific equipment like electric vehicles (EVs), solar panels (PVs) or batteries can be identified which allows utility representatives to offer better tariffs to electricity consumers and such activity can also lead to improved data quality on the utility side.

The ability to perform energy disaggregation analytics holds significant importance for utilities and electricity distribution. By gaining granular insights into customers' energy usage at the appliance level, utilities can develop targeted demand response programs, optimize load distribution, and enhance grid management. Energy disaggregation analytics enables utilities to identify peak demand periods, forecast load patterns, and make informed decisions regarding infrastructure investments. As utilities strive for more efficient and sustainable energy distribution, the ability to perform energy disaggregation analytics becomes increasingly vital in driving operational excellence and meeting the evolving needs of customers and the grid.

The Data Solutions and Methods

To perform energy disaggregation analytics, data analysts typically need the following data:

1. End consumer consumption data - this is the load/consumption profile data of the building or household. It can be obtained from smart meters, utility bills, or energy monitoring systems. The higher the temporal resolution of the data, the better (e.g., minute-level or second-level) to capture detailed appliance usage patterns.

2. End consumer information - additional information from the Client Information System (CIS) which provides valuable information like type of consumer, tariff, geolocation, etc.
3. Time-synchronized data - to accurately correlate energy consumption profiles with specific appliance activities, it's important to have time-synchronized data. This means having data that aligns the timestamps of the energy consumption readings with the corresponding appliance-level data.
4. Additional data that can enhance the accuracy and performance of energy disaggregation analytics include:
 - a. Weather data - weather conditions can influence the energy consumption of certain appliances, such as heating or cooling systems. Therefore, integrating weather data, such as temperature, humidity, or solar radiation, can help improve the accuracy of disaggregation algorithms.
 - b. Occupancy data - knowing the occupancy patterns of a building or household can provide valuable context for energy disaggregation. Occupancy data can be obtained from occupancy sensors, smart home systems, or manual surveys.

The Awesense Energy Transition Platform supports all the above-mentioned data and can ingest, synchronize, centralize and conveniently serve all of them. The serve portion is secured through a comprehensive suite of API calls - REST and SQL APIs through Awesense's Energy Data Model (EDM).

It's important to note that the availability and quality of data play a significant role in the accuracy and effectiveness of energy disaggregation analytics. Higher-resolution and more comprehensive data sets generally yield better results. For these reasons, Awesense Platform offers built-in Validation Estimation Editing (VEE) engine which significantly improves the quality of the ingested data.

There are various techniques used for energy disaggregation, including rule-based methods, statistical analysis, and machine learning algorithms such as Hidden Markov Models (HMMs), Support Vector Machines (SVMs), or deep learning models like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). The choice of technique depends on the available data, computational resources, and desired accuracy.

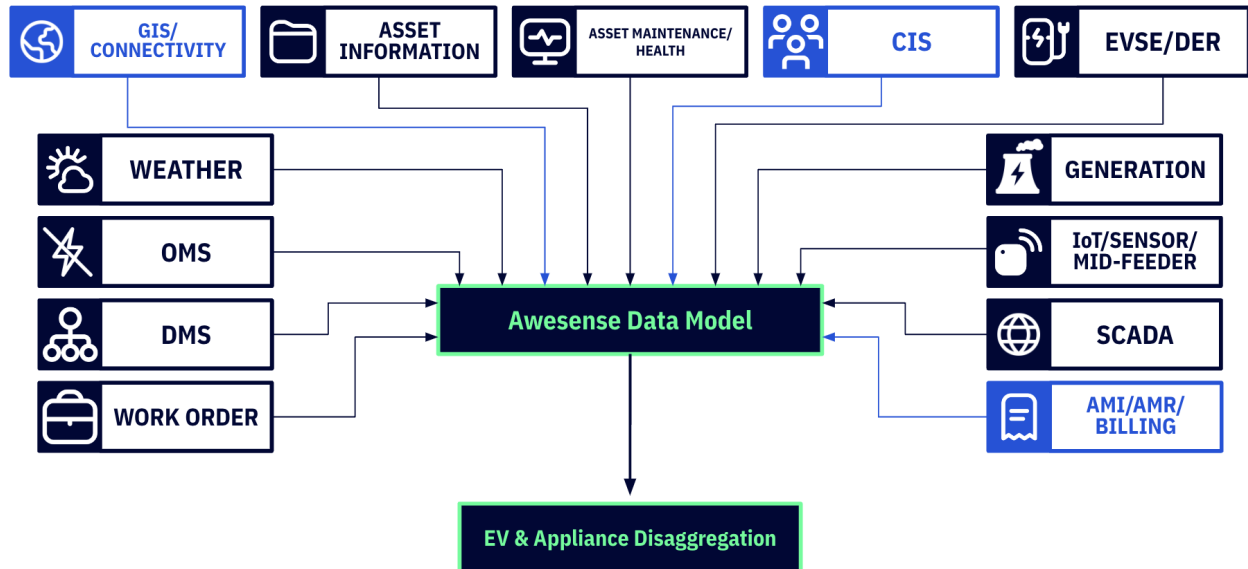
What is important to mention is that the Awesense Platform can conveniently serve data needed for energy disaggregation without limitation on analysis techniques or tool/technology used.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.

EV & Appliance Disaggregation

[UC34-01] - Awesense Energy Transition Platform Use Case



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. A GitHub link to the Jupyter Notebook can be provided upon request.